

Traffic Engineering in Computer Networks (56 Hours)

Course Code	Course Title	Credits	Type (T+P+Pj)
CUDM2511	Traffic Engineering in Computer Networks	3	1+2+0

Prerequisites: Computer Networks, Network Protocols, Basic Probability/Statistics

Course Objectives:

- 1) Understand traffic behavior and modeling in computer networks.
- 2) Learn the principles and techniques of traffic engineering.
- 3) Explore QoS (Quality of Service), MPLS, SDN, and traffic optimization methods.
- 4) Design and simulate traffic-aware networks.

Course Outcome :

CO Code	Course Outcome
CO1	Understand traffic behavior, types, metrics, and basic models in computer networks.
CO2	Analyze and apply QoS techniques and traffic engineering mechanisms like MPLS and constraint-based routing.
CO3	Design and simulate SDN-based traffic-aware networks using real-time path computation and dynamic routing.
CO4	Perform traffic analysis, optimization, and simulation using modern tools such as NS-3, Mininet, and Wireshark.

CO-PO-PSO Mapping Matrix

CO \ PO/PSO	P O1	P O2	P O3	P O4	P O5	PO6–PO10	PO1 1	PO1 2	PSO 1	PS O2
CO1	3	2	1	–	–	–	–	2	2	1
CO2	3	3	3	2	2	–	–	–	3	2
CO3	2	2	3	2	3	–	–	2	3	3
CO4	2	3	3	2	3	–	1	2	3	3

Module 1: Introduction to Traffic Engineering

Need for traffic engineering, Goals and benefits, Types of network traffic (data, voice, video), Performance metrics: throughput, delay, jitter, packet loss, Review of network topologies and protocols

Practicals:

- ❖ Analyze and compare data, voice, and video traffic using **iPerf** and **Wireshark**.
- ❖ Measure network performance metrics such as **throughput**, **delay**, **jitter**, and **packet loss**.
- ❖ Simulate basic network topologies (star, mesh, client-server) using **NS-3** or **GNS3** to observe traffic behavior.

Module 2: Traffic Models and Characterization

Deterministic and stochastic traffic models, Poisson, Pareto, Markov models, Self-similarity and long-range dependence, Measurement tools and techniques

Practicals:

- ❖ Model traffic using **Poisson** and **Pareto** distributions in **NS-3** and **Python (NumPy + Matplotlib)**.
- ❖ Analyze **self-similarity** and **long-range dependence** using trace data from **D-ITG** or **R**.
- ❖ Visualize traffic burstiness and packet arrival patterns to understand real-world traffic behavior.

Module 3: Quality of Service (QoS) and Resource Management

QoS parameters and architecture, QoS in IP networks: IntServ and DiffServ

, Admission control, scheduling, traffic shaping and policing, Weighted Fair Queuing (WFQ), Token Bucket, Leaky Bucket

Practicals :

- ❖ Simulate **Token Bucket** and **Leaky Bucket** algorithms using **NS-3**.
- ❖ Configure **IntServ** and **DiffServ** QoS architectures in **GNS3** or **Mininet**, and evaluate effects on VoIP and HTTP traffic.
- ❖ Implement and compare traffic **scheduling algorithms** such as **FIFO** and **Weighted Fair Queuing (WFQ)**.

Module 4: MPLS and Traffic Engineering

MPLS architecture and components, Label Switched Paths (LSPs), Constraint-Based Routing (CBR), RSVP-TE and CR-LDP, Traffic engineering with MPLS

Practicals:

- ❖ Set up **MPLS routing** with Label Switched Paths (LSPs) using **GNS3** or **Cisco Packet Tracer**.
- ❖ Implement **Constraint-Based Routing (CBR)** using **RSVP-TE** or **CR-LDP** in **NS-3** or **Mininet**.
- ❖ Monitor label forwarding and analyze path selection based on bandwidth and delay constraints.

Module 5: Software-Defined Networking (SDN) and TE

Introduction to SDN, OpenFlow and controller-based TE, Path computation algorithms (Dijkstra, CSPF), Load balancing and dynamic rerouting

Practicals :

- ❖ Design an **SDN network** using **Mininet** and program flow control using **OpenFlow** with **POX/Ryu controllers**.
- ❖ Implement **Dijkstra** and **CSPF** algorithms in **Python** for traffic path computation.
- ❖ Simulate **dynamic rerouting** and **load balancing** by adjusting network conditions in real time.

Module 6: Traffic Analysis, Optimization & Simulation Tools

Network calculus, Optimization techniques (linear, convex) Simulation tools: NS-3, Mininet, GNS3, Real-world case studies: ISP backbones, data centers

Practicals:

- ❖ Capture and analyze live traffic using **Wireshark**, **NetFlow**, and **sFlow**.
- ❖ Use **Python (CVXPY or PuLP)** to formulate and solve traffic **optimization problems** (e.g., delay minimization).
- ❖ Design and simulate a complete **traffic-engineered network (e.g., ISP/campus)** using **NS-3**, **Mininet**, and SDN/MPLS integration in a **mini-project**.

Assessment Methods:

- ❖ Midterm and Final Exam
- ❖ Network simulation lab assignments
- ❖ Case study report or mini-project
- ❖ Quizzes and tutorials

Suggested Tools & Labs:

- i. **Simulation tools:** NS-3, OMNeT++, Mininet, GNS3
- ii. **Traffic generators:** iPerf, D-ITG, Ostinato
- iii. **Analysis tools:** Wireshark, NetFlow, sFlow

Recommended Textbooks & References:

- I. *Traffic Engineering with MPLS* – Eric Osborne & Ajay Simha
- II. *Quality of Service in IP Networks* – Srinivas Vegesna
- III. *Computer Networks* – Andrew S. Tanenbaum, David Wetherall
- IV. Research papers from IEEE/ACM on traffic engineering & SDN